MILITARY SPECIFICATION

BRAIDED FUZE MATERIAL

1. SCOPE

1.1 This specification covers a bimetallic fuze wire braided fuze material.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Fuze material shall be in wire form - 4M8SB (eight strands of 0.004 inch diameter (each strand), high resistance pyrofuze wire coaxially braided in an interlocking configuration.).

FSC 1390

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3.2 Construction. Fuze material shall consist of a metallic shell over a metallic core with net round wire configuration as specified in 3.1.

3.2.1 Shell. The shell shall conform to the following chemical composition:

Ruthenium[1] - 5.0 ± 0.5 percent
Palladium[1] - Remainder

[1]The residual platinum group elements are to be counted as palladium.

3.2.2 Core. The core shall be 5056 aluminum and shall conform to the following chemical composition by weight.

Silicon, percent, max.	0.30
Iron, percent, max.	0.40
Copper, percent, max.	0.10
Zinc, percent, max.	0.10
Manganese, percent	0.05 - 0.20
Magnesium, percent	4.5 - 5.6
Chromium, percent	0.05 - 0.20
Aluminum, percent	Remainder

Note. The contractor shall furnish a certificate stating that the chemical composition of the braided wire conforms to the chemical requirements of this specification.

3.3 Density. The single strand wire from which the braid is fabricated shall have the following ratio: milligrams/200 millimeters of wire divided by diameter of the wire in mils squared = 0.71 to 0.79*.

*The value of 0.71 to 0.79 is defined as the proportional factor for single strand wire.

3.4 Weight per unit length. A foot length of braided fuze material shall weigh 146 ± 5 milligrams.

3.5 Breaking load. The breaking load of the braided fuze (4M8SB) shall be 9.5 pounds minimum.

3.6 Burning rate. The braided fuze material shall have a burning rate of 1300 milliseconds per linear foot at ambient temperature.

3.7 Electrical resistance. A single strand of wire shall have an equivalent electrical resistance for its diameter consistent with a bulk resistivity of 62 ± 2 ohms per circular mil foot at 20°C.

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3.8 Performance. When tested as indicated in 4.6.7, material shall undergo a linear disintegration into minute discontinuous globules resulting in the release of approximately 320 calories of heat per gram of reacted material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will insure equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.8 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lot. A lot shall consist of the fuze material produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions in one unchanged process. However, the ingredients shall be the product of one manufacturer.

4.4 Sampling for examination and tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-4. A five foot sample of braided fuze wire shall be taken from each spool of wire at least three feet from the free end, labeled clearly for identification as to the type of material and the lot represented.

4.5 Inspection provisions.

4.5.1 For examination and nondestructive tests. Sample wire shall be examined and tested in accordance with 4.6 and MIL-STD-105. The acceptance number is zero.

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4.6 Test procedures.

4.6.1 Material. Laboratory tests shall be performed to determine the composition of 3.2.1 and 3.2.2 by approved analytical methods.

4.6.2 Density. Select a convenient length of strand, about 200 millimeters, measure its length to the nearest millimeter, and weigh accurately. Determine the diameter with a micrometer to nearest 0.0001 inch. Calculate the density using the following formula:

$$\text{Density} = \frac{W/L}{D^2}$$

where W = weight in milligrams
L = length in millimeters
D = diameter (mils)

Make three determinations from different spools and report as an average.

4.6.3 Weight per unit. Test specimens shall be selected at least 3 feet from the end of a spool. The braided wire shall be drawn from the spool under low even tension so that no elongation of the wire takes place. Each test specimen shall be cut to a length of one foot. The test specimen shall be folded upon itself several times and twisted to make a compact bundle. The operator's hands shall be clean and dry. The balance shall be handled with extreme care when actual weighings are being made. At least three weighing shall be made.

4.6.4 Tensile strength. Tensile strength shall be determined using standard measuring equipment of a pendulum type tensile machine reading to an accuracy of at least 5 grams and capable of applying force at a uniform rate of 12 inches per minute. Make five determinations from different spools or one determination from each spool if sample size is less than five. Report results as an average.

4.6.5 Burning rate. From each lot, a test sample of wire approximately 12 inch long shall be taken. Use a test fixture consisting of a clamping device capable of holding a ten (10) inch length of the specimen horizontally an electrical contact at each end of the 10 inch specimen, and a start-stop chronograph for timing the burning rate. At ambient temperature, the burning rate of the samples tested shall be 1 second plus or minus 5 percent when measured at one atmosphere of air pressure at 70°F.

4.6.6 Electrical resistance. Electrical resistance shall be determined by means of a Wheatstone bridge. Resistance shall be reported as the average value determined from not less than five measurements on one-foot lengths of braided wire selected from the sample.

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4.6.7 Performance. Performance shall be determined using a calorimeter bomb. Five determinations shall be made from specimen taken from the sample. Report results as an average.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level C. Unless otherwise specified, the wire shall be furnished wound upon a spool. The quantity of wire per spool shall be in accordance with standard commercial practice.

5.2 Packing.

5.2.1 Level C. Spools shall be packed in containers in such a manner as to insure carrier acceptance and safe delivery to the point of destination.

5.3 Marking. Each spool, whether packed in a container, bunbled, or shipped, shall carry an identification tag or label on which is clearly stamped or indelibly printed, the date of manufacture, dimension, customer order number, quantity of feet, material designation, and bar number or numbers.

5.3.1 In addition to any special marking required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This material is intended to be used as a fuze wire which will actuate when externally heated at a rapid rate. The material will perform in any atmosphere.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity of material.
(c) Place of inspection

6.3 All wire destroyed or consumed in conducting the tests required by this specification should be in addition to the quantities of wire ordered and the cost thereof should be included in the contract or order.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1390-A438

Review interest:

Army - MU

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES
NAVY DEPARTME

OFFICIAL BUSINESS

COMMANDING OFFICER
PICATINNY ARSENAL
ATTN: SMUPA-DC-7
DOVER, NEW JERSEY

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PLATE NO. 15419

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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